LN Group 15 Report

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1 Introduction

In this report we present the result of 3 models that classify rows of a medical dataset by the corresponding medical speciality. After building a preliminary rule-based model (Section 3.1), we asked: can a model *without* the keyword matcher (not exploiting an existing data-leak) and, if so, which of their properties make this possible?

2 Data

The dataset contains medical records with five fields: **medical_specialty**, the target label, one of 12 specialties (e.g. *Surgery*, *Orthopedic*), absent from the test set; **description**, a short summary of the complaint or procedure (e.g. *"Chest pain, possible syncopal spells."*); **sample_name**, a short title (e.g. *"Chest Pain - Cardiac Consult"*); **transcription**, the full clinical note, with subsections such as *HISTORY OF PRESENT ILLNESS...*; and **keywords**, a list of relevant terms (e.g. *cardiovascular-pulmonary, chest pain, syncopal, . . .*). **In most records of the training rows, the first keyword is the medical_specialty itself (data leak).**

2.1 Data Cleaning

In every model we used the same procedures for cleaning the data, namely: text normalization, removing stop words, removing punctuation.

In order to do some keyword selection and avoid diluting their informative potential, we remove keywords with a very low cross-entropy, that is, keywords that appear in most of the labels, as they do not help to distinguish between different labels.

2.2 Data Analysis

Two points about keywords and class imbalance: (i) **keywords** condense each note and usually **start with the row's own class** (a *leak* we exploit), but some rows (≈20%) lack it, requiring a fallback; and (ii) **classes are not balanced** in the training set: **dominant:** Surgery (33.5%); **medium:** Cardiovascular-Pulmonary (11.4%), Orthopedic (11.2%), Radiology (8.3%), General Medicine (7.9%), Gastroenterology (7.3%), Neurology (6.9%), Obstetrics-Gynecology (5.2%); **minority:** Neurosurgery (3.0%), Ophthalmology (2.7%), Psychiatry-Psychology (1.5%), Dermatology (1.0%).

2.3 Data Augmentation (DA)

We acknowledge that the DA method implemented in our project might instead be considered a data cleaning technique. However, we justify our method as DA as it also fits with characteristics of distant supervision. [noa,] [Feng et al., 2021]

2.3.1 Keywords Augmentation

Keywords hold most of the information needed for classification and are simple to tokenise, but 444

of the 2616 training rows (17%) and 78 of the 398 test rows (20%) have none. Without requiring medical knowledge, we built a global keyword set from the training set (excluding stop words, numbers and single characters, and the 5% of keywords with the highest entropy across classes, which do not discriminate between specialties). Each row with no keywords then receives, as keywords, the words of its other columns that belong to this set; rows that already have keywords are not changed. As more data is added this set grows, and with large datasets it may need further pruning to keep its information density.

3 Models

3.1 Rule-Based (nMED) + fallback

Exploiting the mentioned *data-leak*, we implemented a rule-based model that uses the normalized Minimum Edit Distance (nMED) in order to match keywords with their respective class. After some iteration, we chose to accept classifications where the keyword had a nMED of 0.90 or more. If it is unable to match, the fallback is the most common class: *surgery*.

3.2 Logistic Regression

From the knowledge gained from implementing 3.1, we chose to implement a Logistic Regression classifier. This seemed the next step as it could beat the naive implementation by incorporating many features instead of a single best match. The question will be if this model will have its classification potential diluted by having more features, or if it will gain new information from the larger feature set.

The first step is to choose the input. We decided to encode the keywords using one-hot vectors for our input. An L2 regularization was chosen, as L1 regularization had worse results, most likely due to collapsing most of the regression weights to 0 and keeping only the weights of the *leaked* keywords, resulting in model closer to the Rule-Based model (3.1). The "balanced" mode adjusted weights inversely proportional to class frequencies in the input data [scikit-learn developers, 2026] [Pedregosa et al., 2011], which greatly improved our models performance.

3.3 Pre-trained Embeddings + Logistic Regression

To isolate the contribution of the keywords (the leak), the third model ignores them and reads only the clinical report (*description*, *sample_name* and *transcription*). If it matched Section 3.2, the keywords would not be required; otherwise, the previous gains come mostly from the leak.

Each report is encoded by a pre-trained BERT-family encoder [Devlin et al., 2019] with `sentence-transformers` [Reimers and Gurevych, 2019][1]. As transcriptions exceed the encoder input limit, each report is split into 250-word chunks, each chunk is encoded into an L2-normalised 768-dimensional vector, and the report vector is their mean. The same classifier as in Section 3.2 is trained on these vectors, so that differences between the two models come only from the input.

We compared clinical (Bio_ClinicalBERT [Alsentzer et al., 2019][2]), biomedical (BiomedBERT [Gu et al., 2020][3], BioBERT [Lee et al., 2020][4], S-PubMedBert[5]) and general-domain (all-mpnet-base-v2[6]) encoders, first individually and then as an ensemble, to

[1] https://www.sbert.net

[2] https://huggingface.co/emilyalsentzer/Bio_ClinicalBERT

[3] https://huggingface.co/microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext

[4] https://huggingface.co/dmis-lab/biobert-base-cased-v1.2

[5] https://huggingface.co/pritamdeka/S-PubMedBert-MS-MARCO

[6] https://huggingface.co/sentence-transformers/all-mpnet-base-v2

test whether encoders pre-trained on different corpora compensate for each other's weaknesses on specific specialties: their class probabilities are combined by the *mean* (soft voting) or the *max* (the most confident encoder decides).

4 Experimental Setup

All models are scored on the training set with accuracy and macro-F1. The rule-based baseline learns nothing, so it is evaluated on the full set and the two Logistic Regression models use 5-fold stratified CV (seed$= 0$). **Hyperparameters: Keyword augmentation**: $DROP_FRAC = 0.05$, the fraction of keywords with the highest entropy across classes (spread evenly over all specialties) removed from the keyword set. **Rule-Based**: $MED_THRESHOLD = 0.90$, the minimum nMED similarity to accept a keyword as a class name; below it, the fallback (Surgery) is used. **LR (keywords)**: $C = 0.015$, the inverse of the L2 regularisation strength; **balanced=True**: balanced class weights in log-regression. **Embeddings + LR**: chunks of 250 words (200 for all-mpnet-base-v2), below the encoder input limit of 512 tokens (384); $C = 50$, chosen on Bio_ClinicalBERT among $\{0.015, 0.1, 1, 10, 50, 100\}$, where macro-F1 rose from 0.32 to a peak of 0.54 at $C = 50$ (0.53 at $C = 100$), and kept for all encoders.

5 Results

The best model is the **Logistic Regression on one-hot keywords (see 3.2)**: 91.8% accuracy (+6.0 pp over the baseline), 92.1% macro-recall (+13.4 pp), 90.3% macro-F1 (+5.8 pp) and a lowest class F1 of 82.6% (+33.5 pp); its confusion matrix is in Appendix 6.

Without keywords, the best encoder was BiomedBERT, with 51.4% accuracy and 53.5% macro-F1 (lowest: Dermatology, 34.8%), 40.4 pp below the keyword-based LR; the other encoders ranged down to 45.4% (all-mpnet-base-v2). The ensemble did not help: the mean raised macro-F1 by only 0.9 pp (54.4%) while lowering accuracy to 50.3%, and the max lowered both (49.1%, 52.3%), suggesting that the encoders make largely the same errors.

6 Discussion & Future Work

Error source 1: rows without the leaked keyword. In 78.5% of the training rows (2051/2613) the first keyword is the class name. On these rows the keyword Logistic Regression (LR) reaches 99.3% accuracy, against 64.4% on the remaining 562 rows which are 93% of its errors (e.g. *"History & physical for right ankle sprain"*, keywords: *sounds, systems, chest, supple, physical, . . .*; *Orthopedic*, predicted: *General Medicine*).

Error source 2: surgery is a procedure of many medical specialities. This created problems in the classification of rows with surgeries in other specialities (e.g. *"Left heart catheterization, left ventriculography, selective coronary angiography"*; *Surgery*, predicted: *Cardiovascular-Pulmonary*).

We conclude that any model that uses the data-leak will have a large advantage compared to the other models. None of the theoretically more powerful BERT-based models was able to achieve the accuracy or F1-score of the rule-based baseline nor the log-regression model. The log-regression model is the best model as it incorporates the leak in its weights while not excluding other possibly relevant features (*keywords*) while also being able to attenuate the problem with the unbalanced classes with its implementation and parameters.

Future work could further study the chosen hyper-parameters and test how bigger deep-learning models compare in this task. Comparing our results with a state-of-the-art LLM could also be an interesting future project.

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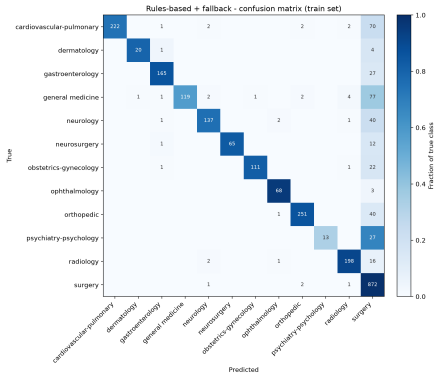
[Pedregosa et al., 2011] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

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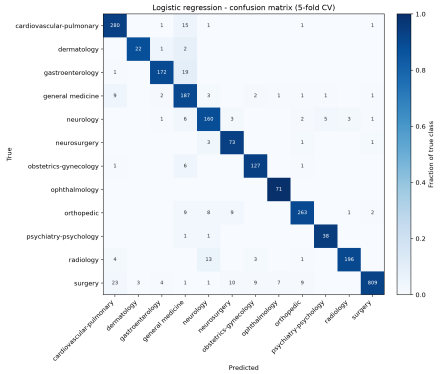
[scikit-learn developers, 2026] scikit-learn developers (2026). *sklearn.linear_model.LogisticRegression*. scikit-learn, version 1.9.1 edition. Accessed: 2026-10-10.

Use of LLMs. We used Claude (Anthropic) to help with the Latex formatting and, in the code, for grid-searching ideal parameters, for overall syntax help and with the choice of the embeddings.

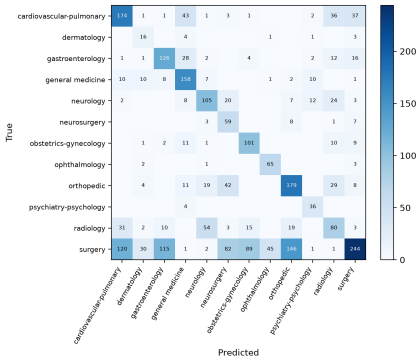
Appendix A: Extra Figures and Tables



(a) Rules-based + fallback confusion matrix. Note the fallback errors in surgery



(b) Logistic Regression confusion matrix. Note the density in the correct predictions and the lack of any clear pattern in errors.



(c) BiomedBERT embeddings + LR confusion matrix. Note the *surgery* row: only 244 of its 876 rows are correct, the rest being spread over the specialties whose procedures they describe

Figure 1: Results for the three approaches.